Aluffi Chapter-1 Solutions

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Exercise 0.1. *Alu09, Ex-1.6 Define a relation $\sim$ on $\mathbb{R}$ as $a \sim b$ if $a - b \in \mathbb{Z}$. Show that $\sim$ is an equivalence relation on $\mathbb{R}$. Find a compelling description for $\mathbb{R}/\sim$. Do the same for the relation $\equiv$ on the plane $\mathbb{R} \times \mathbb{R}$ defined by $(a_1, a_2) \equiv (b_1, b_2) \Leftrightarrow b_1 - a_1 \in \mathbb{Z}$ and $b_2 - a_2 \in \mathbb{Z}$.*

Proof. Reflexivity holds because $x - x = 0 \in \mathbb{Z}$. Symmetry is due to $y - x = -(x - y) \in \mathbb{Z}$. Finally, for $x, y, z \in \mathbb{R}$ with $x-y = a \in \mathbb{Z}$ and $y-z = b \in \mathbb{Z}$, we can write $x-z = x-y-(z-y) = a+b \in \mathbb{Z}$ which establishes transitivity. This completes the porof that $\sim$ is an equivalence relation.

Next, we will show that $\mathbb{R}/\sim$ is in bijection to the following set:

$$S^1 := \{x \in \mathbb{C}; \|x\| = 1\}$$

For this we introduce a map

$$g : \mathbb{R} \to S^1$$

defined by

$$x \mapsto e^{2\pi i x}$$

That g is well defined is obvious from intro analysis. We know from intro analysis that g is a surjection.

Then, we shall invoke Alu09, Thm-2.7 on the diagram

$$\begin{tikzcd}
\mathbb{R} \arrow[r, "g"] \arrow[d, "p"'] & S^1 \\
\mathbb{R}/\sim \arrow[ur, "f"'] &
\end{tikzcd}$$

This gives us a well defined injection $f : \mathbb{R}/\sim \to S^1$. We get that $\operatorname{im} f = \operatorname{im} g$ and since g is surjective $\operatorname{im} f = S^1$ which completes the proof.

Next, we make an important observation.

Observe. *Since any pair of points in any set in $\mathbb{R}/\sim$ have the same fractional part, quotienting $\mathbb{R}$ by $\sim$ has the effect of treating the integral part of any real as insignificant and of course, those reals with identical fractional parts are identified as equivalent. Motivated by this, hereafter, we will simply use the notation $\mathbb{R}/\mathbb{Z}$ for $\mathbb{R}/\sim$.*

Since $\mathbb{R}/\mathbb{Z} \cong S^1$ we are tempted to believe that $\mathbb{R}^2/\mathbb{Z}^2$ is in bijection with $S^1 \times S^1$. We shall prove this.

First, notice that there is a bijection $f : \mathbb{R}/\mathbb{Z} \to S^1$. This gives us a bijection

$$\mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z} \to S^1 \times S^1$$

by the rule

$$(x + \mathbb{Z}, y + \mathbb{Z}) \mapsto (f(x), f(y))$$

So, we simply need to prove that there is a bijection between $\mathbb{R}^2/\mathbb{Z}^2$ and $\mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z}$.

Here, we shall use the universal property of products as in Kim18, Prop-1.3.1.
For this, consider the function

$$f : \mathbb{R}^2/\mathbb{Z}^2 \to \mathbb{R}/\mathbb{Z}$$

defined by

$$(x, y) + \mathbb{Z} \times \mathbb{Z} \mapsto x + \mathbb{Z}$$

and the function

$$g : \mathbb{R}^2/\mathbb{Z}^2 \to \mathbb{R}/\mathbb{Z}$$

defined by

$$(x, y) + \mathbb{Z} \times \mathbb{Z} \mapsto y + \mathbb{Z}$$

Next, consider the diagram

$$\begin{tikzcd}
 & \mathbb{R}^2/\mathbb{Z}^2 \arrow[dl, "f"'] \arrow[d, dashed, "h"'] \arrow[dr, "g"] & \\
 & \mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z} \arrow[dl, "\pi_1"] \arrow[dr, "\pi_2"'] & \\
\mathbb{R}/\mathbb{Z} & & \mathbb{R}/\mathbb{Z}
\end{tikzcd}$$

By universal properties of products, there is a map

$$h : \mathbb{R}^2/\mathbb{Z}^2 \to \mathbb{R}/\mathbb{Z} \times \mathbb{R}/\mathbb{Z}$$

which satisfies

$$f = \pi_1 \circ h, \qquad g = \pi_2 \circ h$$

It is obvious that f and g are surjective and from this follows the surjectivity of h.
Next, take any

$$(x, y) + \mathbb{Z}^2 \in \mathbb{R}^2/\mathbb{Z}^2$$

and

$$(x', y') + \mathbb{Z}^2 \in \mathbb{R}^2/\mathbb{Z}^2$$

so that $h\left((x, y) + \mathbb{Z}^2\right) = h\left((x', y') + \mathbb{Z}^2\right)$
Then, $(\pi_1 \circ h)((x, y) + \mathbb{Z}^2) = x + \mathbb{Z}$ and $(\pi_2 \circ h)((x, y) + \mathbb{Z}^2) = y + \mathbb{Z}$ and a similar result holds for $(x', y') + \mathbb{Z}^2$.
So, if

$$(x + \mathbb{Z}, y + \mathbb{Z}) = (x' + \mathbb{Z}, y' + \mathbb{Z})$$

then we have

$$x + \mathbb{Z} = x' + \mathbb{Z}, \qquad y + \mathbb{Z} = y' + \mathbb{Z}$$

This tells us that $x \equiv x' \pmod{\mathbb{Z}}$ and $y \equiv y' \pmod{\mathbb{Z}}$. This is precisely the condition for

$$(x, y) \equiv (x', y') \pmod{\mathbb{Z}^2}$$

which is another way of saying that $(x, y)+\mathbb{Z}^2 = (x', y')+\mathbb{Z}^2$ which proves that h is an injection thereby completing the proof. $\square$

Remark. It is worth mentioning that the second half of 0.1 could be proved by invoking the universal property of the quotient $\mathbb{R}^2/\mathbb{Z}^2$ instead of invoking the universal property of the product $\mathbb{R}/\mathbb{Z}\times\mathbb{R}/\mathbb{Z}$.

In fact, we can combine the two universal properties together. We can use the property on $\mathbb{R}^2/\mathbb{Z}^2$ to find a unique map s that is injective and realize that s is the same as h by invoking both universal properties.

$$\begin{tikzcd}
\mathbb{R}^2 \arrow[r, "p"] \arrow[dr, "r"] & \mathbb{R}^2/\mathbb{Z}^2 \arrow[d, dashed, "{h,s}"] \arrow[ddl, "f"'] \arrow[ddr, "g"] & \\
& \mathbb{R}/\mathbb{Z}\times\mathbb{R}/\mathbb{Z} \arrow[dl, "\pi_1"] \arrow[dr, "\pi_2"'] & \\
\mathbb{R}/\mathbb{Z} & & \mathbb{R}/\mathbb{Z}
\end{tikzcd}$$

Here, p is the projection of $\mathbb{R}^2$ onto the quotient $\mathbb{R}^2/\mathbb{Z}^2$, $r : \mathbb{R}^2 \to \mathbb{R}/\mathbb{Z}\times\mathbb{R}/\mathbb{Z}$ is the map $(x, y) \mapsto (x+\mathbb{Z}, y+\mathbb{Z})$.

Exercise 0.2. Alu09, *Prop-2.1.2 Let $f : A \to B$ be a function between any two sets A and B. Prove that f has a right inverse if and only if f is surjective.*

Proof. ($\Longrightarrow$) Suppose $\operatorname{im} f = B$. Then, for any $b \in B$ the set $f^{-1}(\{b\})$ (the fiber of f at b) is nonempty. So, by axiom of choice, for every $b \in B$ we choose a $b^\dagger \in A$ so that $b^\dagger \in f^{-1}(\{b\})$. By this, we define the map

$$\begin{aligned} g\colon B &\longrightarrow A \\ b &\longmapsto b^\dagger \in f^{-1}(\{b\}) \end{aligned}$$

Next, take any $b \in b$. Then, $(f \circ g)(b) = f(g(b)) = f(b^\dagger) = b$. Therefore, $f \circ g = \mathbb{1}_B$.

($\Longleftarrow$) Suppose there is a map $g : B \to A$ so that $f \circ g = \mathbb{1}_B$. Then, pick any $b \in B$. Then, $g(b)$ is an element of A so that

$$f(g(b)) = b \tag{1}$$

proving that $\operatorname{im} f = B$.

$\square$

Remark. Notice that we require no further condition on the sets A and B.

Exercise 0.3. Alu09, *Ex-2.6 Explain how any function $f : A \to B$ determines a section of the projection $\pi_A : A \times B \to A$.*

Proof. The map $g : A \to A \times B$ defined by $a \mapsto (a, f(a))$ is a section of π_A. Indeed, given any $a \in A$,

$$
\begin{aligned}
(\pi_A \circ g)(a) &= \pi_A(g(a)) \\
&= \pi_A((a, f(a)) \\
&= a
\end{aligned}
$$

Therefore, $\pi \circ g = \mathbb{1}_A$.

$\square$

Definition 0.1. A function $f : A \to B$ between any two sets A and B is said to be an *epimorphism* if for every set Z and a pair of maps $\alpha', \ \alpha'' : B \to Z$ it holds that $\alpha' \circ f = \alpha'' \circ f$ then $\alpha' = \alpha''$.

Exercise 0.4. Alu09, *Ex-2.5 A function $f : A \to B$ between any two sets A and B is surjective if and only if it is an epimorphism (0.1).*

Proof. ($\Longrightarrow$) Suppose $f : A \to B$ is surjective. Then, there is a function $g : B \to A$ so that $f \circ g = \mathbb{1}_B$.
Suppose Z is any set and $\alpha', \alpha''\colon B \to Z$ are any pair of functions so that $\alpha' \circ f = \alpha'' \circ f$.
This implies that

$$\alpha' \circ (f \circ g) = \alpha'' \circ (f \circ g)$$

from which we see that

$$\alpha' \circ \mathbb{1}_B = \alpha'' \circ \mathbb{1}_B$$

from which we derive that $\alpha' = \alpha''$. This completes one side of the proof.
($\Longleftarrow$) Now, suppose f is an epimorphism. Set $Z \coloneqq \{0, 1\}$, $\alpha' \coloneqq \mathbb{1}_B$ and $\alpha'' \coloneqq \mathbb{1}_{\operatorname{im} f}$. Notice that α' and α'' agree on the image of f. Therefore, by the fact that f is an epimorphism, they are identical. Since the indicators of two sets are identical, the two sets are equal and this concludes our proof. $\square$

Definition 0.2. A category C consists of

- A class $\mathsf{Obj}(\mathsf{C})$ of *objects* of the category and
- for every pair of objects A, B of C, a set $\mathsf{Hom}_{\mathsf{C}}(A, B)$ of *morphisms*, with the following properties:
  1. For every object A of C (at least) one morphism $1_A \in \mathsf{Hom}_{\mathsf{C}}(A, A)$, called the *identity* on A.
  2. Any two morphisms $f \in \mathsf{Hom}_{\mathsf{C}}(A, B)$ and $g \in \mathsf{Hom}_{\mathsf{C}}(B, C)$ determine another morphism, denoted $gf \in \mathsf{Hom}_{\mathsf{C}}(A, C)$. That is, for every triple of objects A, B and C of C, there is a function (of sets)
  $$\mathsf{Hom}_{\mathsf{C}}(A, B) \times \mathsf{Hom}_{\mathsf{C}}(B, C) \longrightarrow \mathsf{Hom}_{\mathsf{C}}(A, C)$$
  and the image of the pair f, g is denoted gf.
  3. The previous composition law is associative: if $f \in \mathsf{Hom}_{\mathsf{C}}(A, B)$, $g \in \mathsf{Hom}_{\mathsf{C}}(B, C)$ and $h \in \mathsf{Hom}_{\mathsf{C}}(C, D)$, then
  $$(hg)f = h(gf)$$
  4. The identity morphisms are identities with respect to composition, that is, for all $f \in \mathsf{Hom}_{\mathsf{C}}(A, B)$ we have
  $$f1_A = f, \qquad 1_B f = f$$

> A further requirement is that $\mathsf{Hom}_{\mathsf{C}}(A, B)$ and $\mathsf{Hom}_{\mathsf{C}}(C, D)$ have to be disjoint unless $A = C$ and $B = D$.

Notation (Endomorphism)**.** *A morphism* f *of an object* $A \in \mathsf{Obj}(\mathsf{C})$ *of a category* C *to itself is called an* endomorphism*;* $\mathsf{Hom}_{\mathsf{C}}(A, A)$ *is denoted by* $\mathsf{End}_{\mathsf{C}}(A)$

References

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